WASP-10b

R-0825

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_250911` · created 2026-07-28T16:12:55+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460929.8963 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.181 +/- 0.021
Transit depth (Rp/Rs)^2	3.29 +/- 0.77 [%]
Orbital Inclination (inc)	87.67 +/- 1.01
Airmass coefficient 1 (a1)	0.991 +/- 0.024
Airmass coefficient 2 (a2)	0.011 +/- 0.02
Scatter in the residuals of the lightcurve fit is	2.45 %
Best Comparison Star	#1 - [312, 172]
Optimal Aperture	2.51
Optimal Annulus	14.02
Transit Duration (day)	0.0873 +/- 0.0083

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.181 ± 0.021 vs 0.1592 (deviation 1.04σ)
Duration fit vs published	0.0873 d vs 0.0946 d (deviation 0.88σ)
Mid-time O–C	-12.4 min
Depth SNR	8.6
Residual scatter	2.45 %

Light curve

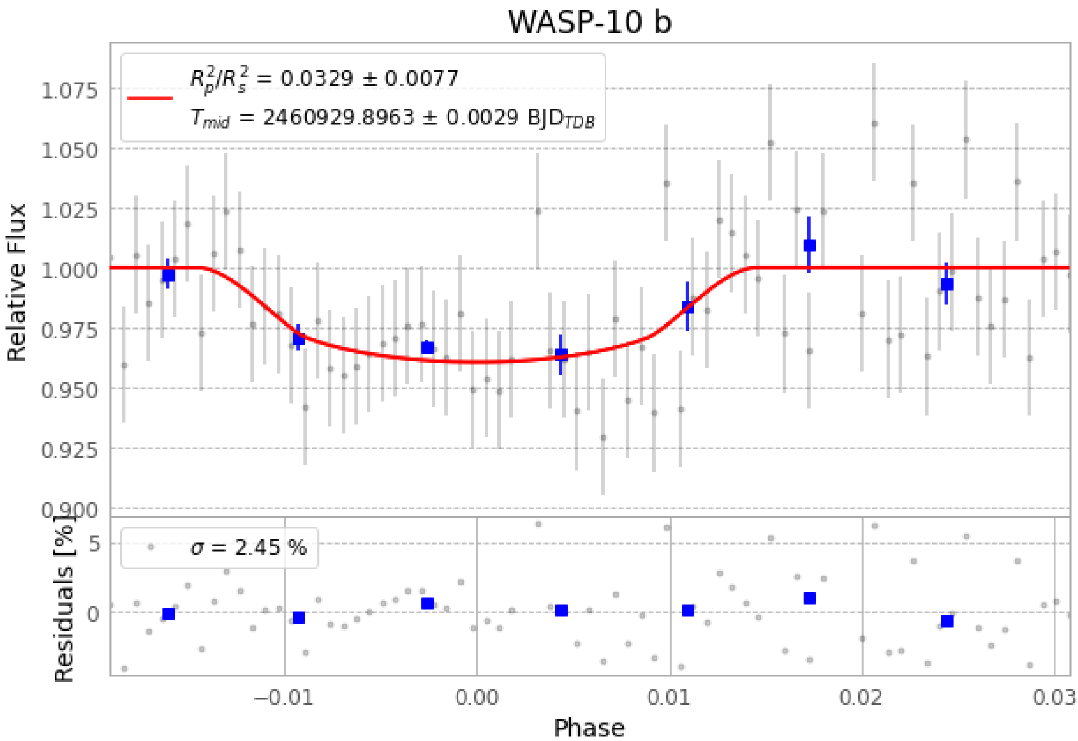

Figure 1 — FinalLightCurve_WASP-10 b_2025-09-11.png (SHA-256 394202e1d3e9d336…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [316, 272], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 4b3a2e75999b0749…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

77 FITS frames (51 MB), `WASP-10250911075715.FITS` → `WASP-10250911114515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-250911.log (72a52552bd67…), FinalLightCurve_WASP-10 b_2025-09-11.png (394202e1d3e9…), FinalLightCurve_WASP-10 b_2025-09-11.csv (2b85d17c1666…)

Compute resources

Wall time 345.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)